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**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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| Office Action Summary | Application No. 17/570,018 | Applicant(s) Fang et al. | |
|---|---|---|---|
| | Examiner KAMAL B DIVECHA | Art Unit 2453 | AIA (First Inventor to File) Status Yes |

*-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --*

**Period for Reply**

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

**Status**

1)☑ Responsive to communication(s) filed on 08/31/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

**Disposition of Claims***

5) ☑ Claim(s) 1-20,23-29,31-32 and 34-37 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20,23-29,31-32 and 34-37 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

**Application Papers**

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

**Priority under 35 U.S.C. § 119**

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

**Attachment(s)**

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 08312026.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

### *Notice of Pre-AIA or AIA Status*

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

**This action is in response to communications filed 08/31/2026.**

**Claims 1-20, 23-29, 31-32, 34-37 are pending.**

### *Information Disclosure Statement*

The information disclosure statement filed **08/31/2026** fails to comply with 37 CFR 1.98(a)(2), which requires a legible copy of each cited foreign patent document; each non-patent literature publication or that portion which caused it to be listed; and all other information or that portion which caused it to be listed. It has been placed in the application file, but the information referred to therein has not been considered.

The document 2973491 EP by Inrix, Inc. is missing from the file wrapper.

### *Response to Arguments*

**Applicant's arguments filed 08/31/2026 have been fully considered but they are not persuasive.**

In the response filed, applicant argues in substance that:

a. Applicant respectfully submits that Han and Shearer, either alone or in combination, do not show, disclose, teach, or suggest at least **"up-sample one or more messages between the first network of the first type and the second network of the second type, wherein: the up-sample inserts additional messages into the one or more messages to generate up-sampled data messages, and the additional messages include interpolations of original values of the one or more messages**," as recited in independent claim 1.

Shearer's interpolated signal samples are not the claimed "additional messages." Claim 1 recites that the CND is structured to "up-sample one or more messages" between the first network and the second network, where "the up-sample inserts additional messages into the one or more messages to

generate up-sampled data messages," and where "the additional messages include interpolations of original values of the one or more messages." Thus, the claimed up- sampling occurs at the message level: additional vehicle-network messages are inserted into a message stream, and those added messages include interpolated values derived from the original message values.

This is not the same in Shearer. Instead, Shearer is on a much lower level. It is directed to increasing packet generation or data rate in a digital communication system by modifying, for example, packets and subcarriers. For example, claim 12 of Shearer recites "[a] system for increasing the data rate of a protocol used for a network transmission" in which a processor receives "a packet" and increases the "standard size of the packet by adding subcarriers to the packet to produce an extended packet." Also, paragraphs [0007] - [0008] of Shearer discuss higher data rates and more effective bandwidth use in WLAN systems. Shearer at paragraph [0035] describes adding subcarriers to a data transmission waveform and multiplying a signal sample rate "(e.g., an interpolation)." Shearer at paragraph [0058] further explains that the interpolation stage "increases the sample rate of the signal," for example, by inserting zeros between original samples and filtering the zero-stuffed sample stream, while "the original spectrum is maintained, and only the bandwidth is doubled." So, Shearer's interpolation is performed on lower-level numerical samples of a physical signal-processing stream, not on higher-level vehicle-network messages, as in claim 1.

Furthermore, at best, Shearer teaches a signal sample that may be interpolated, but this interpolated sample is not an added message, as in the independent claims. Increasing packet size or data rate by adding subcarriers to a packet is not the same as up-sampling messages by inserting additional messages into one or more original messages, as required by independent claim 1.

Therefore, Applicant respectfully submits that Han and Shearer, either alone or in combination, do not show, disclose, teach, or suggest every recitation of independent claim 1, or the similar recitations of amended independent claims 18 and 28 (**remarks, pg. 9-11).**

In response to applicant arguments above, examiner respectfully disagrees.

First, it should be noted that claim 1 recites:

"...up-sample one or more messages...wherein **the up-sample inserts additional** messages **into the one or more messages** to generate up-sampled data messages...**the additional messages include interpolations of original values..."**

**Note: BRI standard is applied.**

**The claim language requires inserting additional messages, which are interpolated values of the original values, into the at least one message (or more is not considered). The claim does not distinguish, nor it defines what the "additional messages" are, i.e. whether they are data packets or data messages in form of data packets, zeroes, values, etc.**

**Applicant above acknowledged that Shearer discloses extending the size of the standard packet by adding subcarriers to the original packet (i.e. message) to produce an extended packet.**

**Furthermore, Shearer also teaches "interpolating each of the plurality of extended packets to produce a plurality of interpolated packets ", e.g. claim 10 and claim 22.**

**In Shearer, in the interpolation stage, the sample rate of the signal is increased, and interpolation is performed by inserting zeros between the original samples, e.g. [0058].**

**Therefore, based on the claim language and BRI standard, Shearer teaches up-sampling at least one message, wherein up-sampling adds additional messages which includes interpolations of the original values, i.e. the extended message is interpolated to generate plurality of interpolated packets.**

**Applicant further argues that "Shearer's interpolations is performed on lower-level numerical samples of a physical signal-processing stream, not on higher-level vehicle-network messages, as in claim 1.**

**However, the examiner would like to emphasize that during patent examination, claims are given their broadest reasonable interpretation and specification is not imported into the claims during patent examination. In this case, claim 1 does not teach or disclose or distinguish between higher level communications and lower-level communications. In fact, in data networks, data is encoded into signals in order to be transmitted via a network. For example: See Shearer [0036-**

**0037]** - Shearer teaches at a lower most layer, the OSI model has a physical layer or PHY layer 102 that is responsible for encoding and **decoding data into signals that are transmitted across a particular medium, e.g. [0036-0037].**

**b. No articulated Reasoning to support the combination (remarks, pg. 11-12).**

The Office Action's proffered explanation to combine Han and Shearer is that "increasing the data rate and allowing more effective use of bandwidth for devices operating in different bands enables more efficient communications and also higher data rate enables service providers to more effectively use their allotted spectrum." See Office Action, page 5. Applicant respectfully submits that this explanation does not provide an articulated reasoning to modify Han's vehicle network gateway to perform the claimed message-level up-sampling.

The Office Action relies on Shearer to teach the claimed "up-sample one or more messages between the first network of the first type and the second network of the second type," where "the up-sample inserts additional messages into the one or more messages to generate up- sampled data messages," and where "the additional messages include interpolations of original values of the one or more messages." See Office Action, page 4.

However, Shearer is not directed towards inserting additional vehicle-network messages into a message stream. **Shearer is directed towards increasing data rate or bandwidth in a digital communication system by modifying packets, subcarriers, waveforms, and signal samples.** For example, claim 12 of Shearer recites "[a] system for increasing the data rate of a protocol used for a network transmission" in which a processor receives "a packet" (which is not a message) and "increase[s] the standard size of the packet by adding subcarriers to the packet to produce an extended packet." Claim 22 of Shearer then recites an "interpolator for increasing the sample rate of each of a plurality of extended packets to produce a plurality of interpolated packets." Similarly, Shearer at paragraph [0035] describes a system that receives digital signals, "multiplies the signal sample rate by the number of received signals (e.g., an interpolation)," shifts the signals with respect to a center frequency, and accumulates the shifted signals into one signal. Shearer at paragraph [0058] further explains that the interpolation stage "increases the sample rate of the signal," for example, by inserting zeros between original samples and low- pass filtering the zero-stuffed sample stream so that the original spectrum is maintained and the bandwidth is doubled. So, Shearer is about low-level signal processing rather than message interpolation.

Accordingly, at best**, Shearer teaches updating packets, physical-layer signal packets, not messages.**

Therefore, **the Office Action's stated motivation does not bridge the technical gap between Han and Shearer.** Increasing data rate or bandwidth in Shearer's WLAN physical-layer signal-processing system does not explain why one of ordinary skill in the art would have modified Han's vehicle gateway to insert additional vehicle-network messages into original vehicle-network messages. Further, the Office Action does not identify objective evidence establishing that Han's gateway would benefit from Shearer's packet/subcarrier/sample-rate interpolation in a manner that would produce the claimed message-level up-sampling

**In response to applicant argument above, examiner respectfully disagrees.**

**First,** In response to applicant's argument that there is no objective evidence establishing that Han's gateway woold benefit from Shearer's data rate solution, the examiner recognizes that obviousness may be established by combining or modifying the teachings of the prior **art to produce the claimed invention where there is some teaching, suggestion, or**

**motivation to do so found either in the references themselves or in the knowledge generally available to one of ordinary skill in the art.** See *In re Fine*, 837 F.2d 1071, 5 USPQ2d 1596 (Fed. Cir. 1988), *In re Jones*, 958 F.2d 347, 21 USPQ2d 1941 (Fed. Cir. 1992), and *KSR International Co. v. Teleflex, Inc.*, 550 U.S. 398, 82 USPQ2d 1385 (2007).

First, here's the **background of applicant's invention:**

BACKGROUND

**[0004]** Vehicle communication networks are utilized to connect sensors, actuators, controllers, and communication devices throughout a vehicle. Recent trends have been increasing the burden on these vehicle communication networks, with more devices being connected, more data passing between devices, lower latency requirements to meet vehicle performance, safety, and emissions requirements, and added vehicle features. Additionally, consumers expect increasing connectivity and features that increase the burdens on vehicle communication networks. These trends are expected to continue, and to accelerate, for the foreseeable future.

**[0005]** Traditional vehicle communication networks (CAN, LIN, FlexRay, MOST, LVDS, etc.) suffer from a number of drawbacks and challenges. These vehicle communication

networks have been developed to meet the particular challenges of a vehicle environment, and have accordingly developed separately from other networks, such as computer local area networks, wide area networks, massively interconnected networks (e.g., the internet), and wireless networks. Most vehicle networks consist of a data link layer and an application layer, utilizing robust and dedicated equipment such as a Controller Area Network (CAN) bus, with dedicated or shared wiring between devices utilizing specific data protocols (e.g., J1939, OBD, etc.). A modern vehicle may have multiple network buses, with specific commands and communications available, and limited customization and data speed available. E.g., CAN buses typically operate at up to about 1 Mbps, with high capability CAN buses operating up to about 10 Mbps. Additionally, CAN buses experience latency greater than 25 ms, and generally higher from about 60 ms to 500 ms, depending upon the configuration, the traffic on the CAN, the priority for particular messages, and the like.

**[0006]** As the number of devices and the data rate demand from the devices increases, traditional vehicle communication networks require the implementation of higher performance buses . Because the automotive industry is a high volume industry with a very low tolerance for failure of components, automotive manufacturers utilize the same components for a long time, and across a broad range of vehicles – including sharing of components across manufacturers. Additionally, a change to a nominally more capable component may introduce risks, integration costs, re-certification burdens for a given application, or have other undesirable consequences to the system. Accordingly, even if vehicle communication networks transition to a higher capability network configuration, it is desirable to keep network types segregated in the system, and to keep a large number of legacy devices (e.g., CAN compatible) in a system for a long period of time.

Secondly, applicant's detailed Description [00103]:

(MOST) network, a Local Interconnect Network (LIN), a FlexRay network, a Time-Triggered Protocol (TTP) network, a Low-Voltage Differential Signaling (LVDS) network, and/or an Ethernet implemented network. In certain embodiments, one or more networks may be an electrical signal zone (e.g., a device providing data and/or receiving commands as an electrical signal, such as a voltage value, a frequency value, and indicated resistance value, or the like), such as a sensor or actuator electrically coupled to an interpreting device that is capable to receive information from, and/or pass information or commands to, one or more electrical devices on the electrical signal zone.

**[00131]** An example system includes the CND 108 interposed between an electrical sensor and one of the networks 406, 410, and structured to provide a sensed value on the network in response to an electrical response of the electrical sensor. For example, one of the networks 406 may be an electrical connection to the second network gateway device 402, with a corresponding end point 408 as the electrical sensor, and whereby the translation circuit 420 converts the electrical signal from the sensor to a communication for the respective network (e.g., network 410, or another network 406). In the example, the translation circuit 420 may perform processing operations on the electrical signal, such as analog/digital (A/D) processing, determination of indicated bits, determination of an indicated value, de-bouncing of the signal, filtering of the signal, diagnostic bit detection (e.g., determination of a fault, and conversion to a corresponding fault value; and/or

**[00157]** Referencing Fig. 18, a schematic depiction of an operation to up-sample a sequence of messages 1802 is depicted. In the example of Fig. 18, a message sequence 1806 (e.g., a series of three communications, in the example) is received, for example, at a network interface circuit of one of the network gateway devices. In the example of Fig. 18, the up-sampling operation is responsive to any up-sampling operations described herein, for example to match a receiving end point data rate, to provide the data represented by the messages 1806 at a scheduled rate, to manage bandwidth on a network of the vehicle and/or for extra vehicle communications, to preserve buffer memory, or for any other purpose, including any up sampling operations of the present disclosure. In the example of Fig. 18, the up-sampling device 1804, which may be a translation circuit, network interface circuit, the CND, a circuit associated with the CND, a circuit regulated by the CND, or the like, generates a translated sequence of messages 1808 (e.g., processed as depicted in Fig. 16 and the related disclosure, and/or according to any other message translation and/or message processing operations set forth herein, and ). The example of Fig. 18 depicts the translated sequence of messages 1808 for clarity of the description. However, the translated sequence of messages 1808 may not all be present at the same time, for example as messages are translated and sent they may be removed, deleted, expire from a cache, etc. The sequence of messages 1808 is depicted to illustrate aspects of the present disclosure. Additionally or alternatively, translation of the messages 1808 may be performed after up-sampling operations are performed, for example to reduce utilization of processing resources.

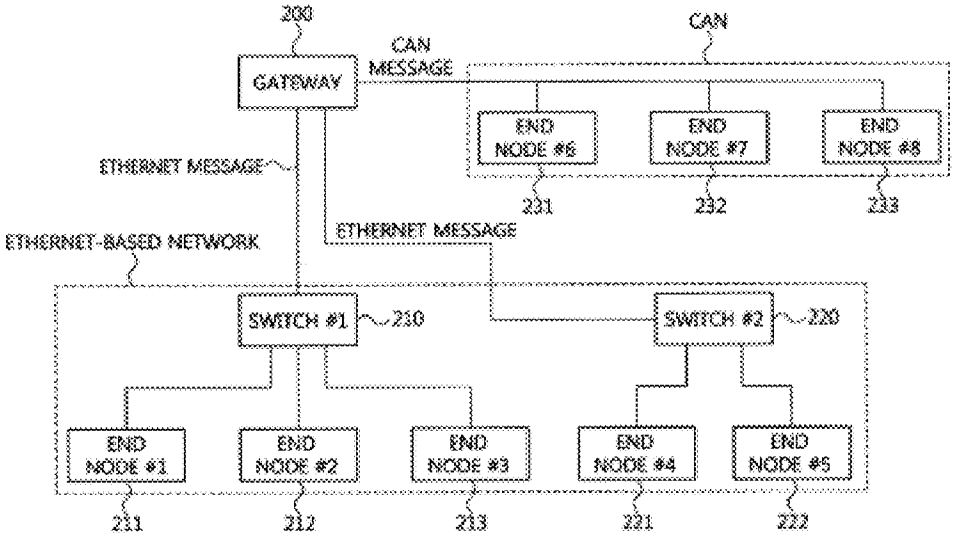
**[00158]** For example, some of the messages may be eliminated or adjusted as a part of the up-sampling before the translation operations (e.g., replacement of frame portions or metadata, encapsulation, processing of the payload and/or frame portions, etc.) are performed. In the example of Fig. 18, an up-sampled sequence of messages 1802 is provided and communicated, for example to a different network gateway device, to a different network of the vehicle from which the first sequence of messages 1806 is received, to an external device (e.g., service tool, cloud server, operator's mobile device, etc.), and/or stored on a

Based on the background and detailed description of the claimed invention, the following points should be noted:

- A need for a higher performance bus **to accommodate legacy vehicle networks that offer low data rates**, bandwidth or capacity and **newer vehicle networks that utilize higher data rates bandwidth or capacity for data communications**.
- **The in-vehicle network is not limited and may comprise wired or wireless connection and/or network with electrical connection to the gateway.**
- **Up-sampling is achieved to MATCH THE RECEIVING ENDPOINT DATA RATE, e.g. see [0157].**
- **Up-sampling is done via interpolation by adding zeroes between the original samples.**
- Applicants' own specification covers **physical layer signal packet processing**.

Han discloses **in-vehicle network comprising a gateway for enabling communications between CAN network and Ethernet network in a vehicle, see fig. 2:**

FIG. 2



Han does not disclose or address the problem associated with different devices with different data rate capabilities.

Shearer discloses the problem associated with different devices with different rates in different networks as follows:

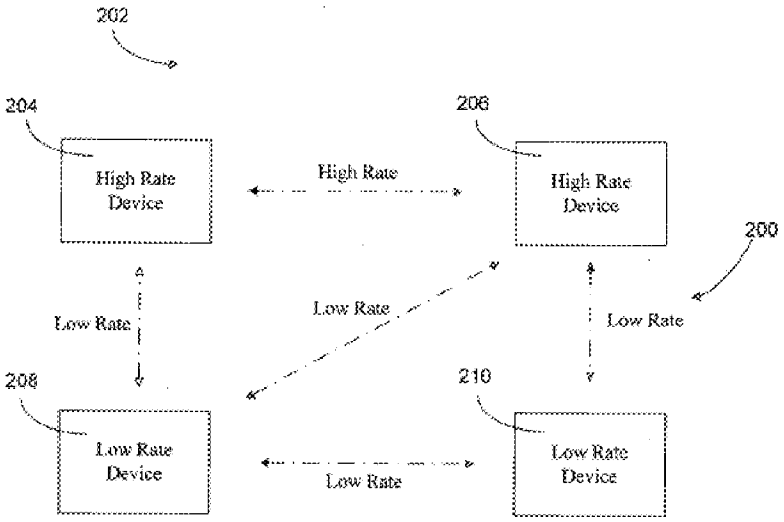



Figure 2

Shearer solves the problem [applicant has demonstrated same problem in the background of the invention, i.e. communication between different devices that supports different data rates, e.g. low and high] **by up-sampling the data sample or packet originated from one device by extending the packet first and then interpolating the extended packet to produce plurality of extended packets, for example: see Shearer [0003-0010].**

**As such, it would have been obvious to a person of ordinary skilled in the art to modify Han in view Shearer in order to increase the data rate, at the gateway device, by up sampling the original data, packets or messages because this would allow more effective use**

**of bandwidth for devices operating in different bands thus enabling more efficient communications.**

[0006] IEEE Standard 802.11 ("802.11") is set out in "Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications" and is available from the IEEE Standards Department, Piscataway, N.J. 802.11 permits either IR or RF communications at 1 Mbps, 2 Mbps and higher data rates, a medium access technique similar to carrier sense multiple access/collision avoidance (CSMA/CA), a power-save mode for battery-operated mobile stations, seamless roaming in a full cellular network, high throughput operation, diverse antenna systems designed to eliminate "dead spots," and an easy interface to existing network infrastructures.

[0007] The 802.11a standard defines data rates of 6, 12, 18, 24, 36 and 54 Mbps in the 5 GHz band. Demand for higher data rates may result in the need for devices that can communicate with each other at the higher rates, yet co-exist in the same WLAN environment or area without significant interference or interruption from each other, regardless of whether the higher data rate devices can communicate with the 802.11a devices. It may further be desired that high data rate devices be able to communicate with the 802.11a devices, such as at any of the standard 802.11a rates.

[0008] Increasing the data rate and allowing more effective use of bandwidth for devices operating in these bands enables more efficient communications. A higher data rate may enable service providers to more effectively use their allotted spectrum. Consumers may realize a cost savings as well.

**The rationale and/or objective evidence supporting the combination can be found in KSR Ruling. See KSR International Co. v. Teleflex Inc.**, 550 U.S. ___,___, 82 USPQ2d 1385, 1395-97 (2007) identified a number of rationales to support a conclusion of obviousness which are consistent with the proper "functional approach" to the determination of obviousness as laid down in Graham. The key to supporting any rejection under 35 U.S.C. 103 is the clear articulation of the reason(s) why the claimed invention would have been obvious. The Supreme Court in KSR noted that the analysis supporting a rejection under 35 U.S.C. 103 should be made explicit, and **MPEP 2143.**

Exemplary rationales that may support a conclusion of obviousness include:
(A) Combining prior art elements according to known methods to yield predictable results;
(B) Simple substitution of one known element for another to obtain predictable results;
(C) Use of known technique to improve similar devices (methods, or products) in the same way;
(D) Applying a known technique to a known device (method, or product) ready for improvement to yield predictable results;
(E) "Obvious to try" – choosing from a finite number of identified, predictable;
(F) Known work in one field of endeavor may prompt variations of it for use in either the same field or a different one based on design incentives or other market forces if the variations are predictable to one of ordinary skill in the art;
(G) Some teaching, suggestion, or motivation in the prior art that would have led one of ordinary skill to modify the prior art reference or to combine prior art reference teachings to arrive at the claimed invention. See MPEP § 2143 for a discussion of the rationales listed above along with examples illustrating how the cited rationales may be used to support a finding of obviousness ].

**Therefore, for the at least these reasons, applicants' arguments are deemed not persuasive.**

***Claim Rejections - 35 USC § 103***

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claim(s) **1-20, 23-26, 28-29, 31-32, 34, 36-37** is/are rejected under **35 U.S.C. 103** as being unpatentable over **Han et al. (hereinafter Han, US 2018/0076970 A1**) in view of **Shearer et al. (US 2006/0018249 A1**).

As per claim 1, Han discloses a system comprising:

a vehicle having a first network of a first type and a second network of a second type, wherein the first type is different from the second type [**fig. 2: Ethernet network connected to gateway which is connected to CAN network, [0050], [0063]];**

a first device on the first network **[fig. 2: [0063]];** and

a converged network device (CND) interposed between the first network and the second network [**fig. 2: Ethernet network connected to gateway which is connected to CAN network, [0050], [0063]];** and structured to:

facilitate communications between the first device and the second network **[fig. 2, 9, 10, [0063], [0100-0112]].**

However, Han does not teach

up-sample one or more messages between the first network of the first type and the second network of the second type,

wherein:

the up-sample inserts additional messages into the one or more messages to generate up-sampled data messages, and

the additional messages include interpolations of original values of the one or more messages.

Shearer, from the same field of endeavor, teaches

up-sample one or more messages between the first network of the first type and the second network of the second type [**claim 12, claim 22],**

wherein:

the up-sample inserts additional messages into the one or more messages to generate up-sampled data messages **[claim 12, claim 22**], and

the additional messages include interpolations of original values of the one or more messages **[claim 12, claim 22, claim 1 and claim 10], [0058], [0035], [0007]].**

Therefore, it would have been obvious to a person of ordinary skilled in the art before the effective filing date of the claimed invention to modify Han in view of Shearer in order to up-sample one or more messages between the first network and second network by interpolations of original values of the one or more original messages.

One of ordinary skilled in the art would have been motivated because increasing the data rate and allowing more effective use of bandwidth for devices operating in different bands enables more efficient communications and also higher data rate enables service providers to more effectively use their allotted spectrum [**Shearer: [0007-0008]].**

As per claim 2, Han-Shearer discloses the system of claim 1, further comprising: a second device on the second network and structured to communicate with the first device via the CND [**Han: fig. 2, 9, 10, [0063-0064], [0100-0112]].**

As per claim 3, Han-Shearer discloses the system of claim 1 further comprising: an external device communicatively coupled to the CND, wherein the external device is structured to adjust a configuration of the CND **[Han: [0088-0091]: 3 modes of operation for the hardware layer. When in sleep mode, it can be waken up when a remote event is detected, i.e. external event from external client or device is received, which changes the configuration mode, inactive, sleep or wake-up of the device**].

As per claim 4, Han-Shearer discloses the system of claim 1, wherein at least one of the first network and/or the second network is a Controller Area Network (CAN) based network [**Han: fig. 2, 9, 10, [0063], [0100-0112]].**

As per claim 5 , Han-Shearer discloses the system of claim 1, wherein at least one of the first network **and/or** the second network is an Ethernet based network **[Han: [fig. 2, 9, 10, [0063-0064], [0100-0112]].**

As per claim 6, Han-Shearer discloses the system of claim 5, wherein the Ethernet based network comprises a data bus architecture **[Han: fig. 1-fig. 2, [0063-0064], [0078]: Bus topology].**

As per claim 7, Han-Shearer discloses the system of claim 1, wherein the first network is a Controller Area Network (CAN) based network and the second network is an Ethernet based network [**Han: fig. 2, 9, 10, [0063-0064], [0100-0112]].**

As per claim 8, Han-Shearer discloses the system of claim 7, wherein the CND comprises:

a communication device structured to access the CAN based network **[Han: Fig. 2 gateway 200, [0063]];**

an Ethernet switch structured to access to the Ethernet based network **[Han: fig. 2 gateway 200, [0093-0098]];** and

wherein the communication device provides a message to the Ethernet switch in response to a corresponding message on the CAN based network **[[0093-0098], fig. 10, fig. 16, [0128-0131]].**

As per claim 9, Han-Shearer discloses the system of claim 8, wherein the communication device is structured to provide the message to the Ethernet switch as an Ethernet message including at least one encapsulated CAN message based on the corresponding message on the CAN based network [**Han:** [**0093-0098], fig. 10, fig. 16, [0128-0131]].**

As per claim 10, Han-Shearer discloses the system of claim 9, wherein the at least one encapsulated CAN message includes more than one encapsulated CAN message [**Han: fig. 3, fig. 14, fig. 15, [0106-0112], [0118]: multiple DU's are included].**

As per claim 11, Han-Shearer discloses the system of claim 8, wherein the communication device is positioned in a first housing, and wherein the Ethernet switch is positioned in a second housing **[Han: fig. 9: CAN PHY layer and Ethernet PHY layer units are separate units/chips, [0080]].**

As per claim 12, Han-Shearer discloses the system of claim 8, wherein the communication device and the Ethernet switch are positioned in a single housing [**Han: fig. 2: Gateway 200, fig. 9, [0080]]**.

As per claim 13, Han-Shearer discloses the system of claim 7, wherein the CND comprises: a communication device structured to access the CAN based network; an Ethernet switch structured to access to the Ethernet based network; and wherein the communication device receives a message from the Ethernet switch in response to a corresponding message on the Ethernet based network [**Han: fig. 2, [0063-0068], fig. 9-10, fig. 16].**

As per claim 14, Han-Shearer discloses the system of claim 13, wherein the communication device is structured to receive the message from the Ethernet switch as an Ethernet message including at least one encapsulated CAN message based on the corresponding message on the Ethernet based network **[Han: [0093-0098], fig. 10, fig. 16, [0128-0131]]**.

As per claim 15, Han-Shearer discloses the system of claim 14, wherein the at least one encapsulated CAN message includes more than one encapsulated CAN message **[Han: fig. 3, fig. 14, fig. 15, [0106-0112], [0118]: multiple DU's are included].**

As per claim 16, Han-Shearer discloses the system of claim 13, wherein the communication device is positioned in a first housing, and wherein the Ethernet switch is positioned in a second housing **[Han: fig. 9: CAN PHY layer and Ethernet PHY layer units are separate units/chips, [0080]]**.

As per claim 17, Han-Shearer discloses the system of claim 13, wherein the communication device and the Ethernet switch are positioned in a single housing **[Han: fig. 2: Gateway 200, fig. 9, [0080]].**

As per claim 18, Han discloses An apparatus comprising:

a first network interface circuit structured to interpret a first network data set having a first network format [**fig. 9-10, fig. 16**];

a translation circuit structured to:

determine one or more message values from the first network data set [**[0106]: payload of the message**] and to encode the one or more message values in a second network data set having a second network format [**fig. 10, [0064-0067] CAN message and Ethernet message, [0094]: conversion, [0109-0111]: generate CAN message from the Ethernet message, fig. 16 and [0127-0131]: generate ethernet message from CAN message**];

second network interface circuit structured to transmit the second network data set [**fig. 2, fig. 9-fig. 10, fig. 16, 0109-0111], 0127-0131];**

wherein:

the first network format and the second network format are vehicle data formats [**fig. 2: Ethernet network connected to gateway which is connected to CAN network, [0050], [0063], [0094]];** and

the first network format is different than the second network format **[fig. 2: Ethernet network connected to gateway which is connected to CAN network, [0050], [0063], [0094]].**

However, Han does not teach

wherein the one or more encoded message values are up-sampled from the one or more message values via an insertion of additional message values into the one or more message values, and the up-sample adjusts a ratio of a number of the one or more encoded messages values to the one or more message values;

the additional message values include interpolations of original values of the one or more message values.

Shearer, from the same field of endeavor, teaches

wherein the one or more encoded message values are up-sampled from the one or more message values via an insertion of additional message values into the one or more message values, and the up-sample adjusts a ratio of a number of the one or more encoded messages values to the one or more message values [**claim 12, claim 22, [0058]],**

the additional message values include interpolations of original values of the one or more message values [**claim 12, claim 22, claim 1 and claim 10], [0058], [0035], [0007**]].

Therefore, it would have been obvious to a person of ordinary skilled in the art before the effective filing date of the claimed invention to modify Han in view of Shearer in order to up-sample one or more messages between the first network and second network by interpolations of original values of the one or more original messages.

One of ordinary skilled in the art would have been motivated because increasing the data rate and allowing more effective use of bandwidth for devices operating in different bands enables more efficient communications and also higher data rate enables service providers to more effectively use their allotted spectrum [**Shearer: [0007-0008]].**

As per claim 19, Han discloses the apparatus of claim 18, further comprising: a configuration circuit structured to configure the first network interface circuit, the translation circuit, and/or the second network interface circuit, responsive to a configuration command value from a device external to a vehicle, the vehicle including the first network interface circuit and the second network interface circuit [**fig. 2, 10, 16, [0088-0091]: 3 modes of operation for the hardware layer. When in sleep mode, it can be waken up when a remote event is detected, i.e. external event from external client or device is received, which changes the configuration mode, inactive, sleep or wake-up of the device, [0067, 0104-0105]: Ethernet message from endpoint in ethernet network includes header, at least one DU. Ethernet message further includes authentication information ASIL such as authentication key, a hash value, CRC, FRC and the like. The header includes destination address DA, source address SA, type filed, IP header, id, flag field, etc. [0129]: CAN message also includes control or management information**].

As per claim 20, Han discloses the apparatus of claim 19, wherein the configuration command value comprises **at least one value** selected **from the values consisting of**: a sampling rate value; a payload processing value; a timestamp value; **a metadata value**; **a target device** (i.e. destination) for the one or more message values from the first network data set, **wherein the target device comprises a device on the vehicle** to **send and/or receive** the

one or more message values; and **an encapsulation description for the one or more message values [[0067, 0104-0105**]: Ethernet message from endpoint in ethernet network includes header, at least one DU. Ethernet message further includes **authentication information ASIL** such as authentication key, a hash value, CRC, FRC and the like. The header includes **destination address DA**, source address SA, **type filed,** IP header, id, flag field, etc. [0129]: CAN message also includes control or management information].

As per claim 23, Han discloses the apparatus of claim 18, wherein one of the first network format or the second network format is Controller Area Network (CAN) based, and the other of the first network format or the second network format is Ethernet based **[Han: fig. 2, 9, 10, [0063], [0094], [0100-0112]].**

As per claim 24, Han discloses the apparatus of claim 23, wherein one or more CAN messages of the CAN based network can be carried in one Ethernet frame of the Ethernet based network [**Han: fig. 2, 9, 10, [0063], [0094], [0100-0112]].**

As per claim 25, Ha discloses the apparatus of claim 18, wherein the first network interface circuit and/or the second network interface circuit is integrated with a configurable edge gateway (CEG) [[**Han: fig. 2, 9, 10, [0063], [0094], [0101-0112]].**

As per claim 26, Ha discloses the apparatus of claim 18, wherein the first network interface circuit and/or the second network interface circuit is integrated with an Ethernet switch [**fig. 2 gateway 200, [0093-0098]].**

As pe claims 28, it does not teach or further define over the limitations in claim 18. Therefore, claim 28 is rejected for the same reasons as set forth in claim 18.

As per claim 29, Han discloses the method of claim 28, further comprising adjusting a configuration at least one of the encoding or the transmitting in response to a configuration change from an external device [**fig. 2, 10, 16, [0088-0091]: 3 modes of operation for the hardware layer. When in sleep mode, it can be waken up when a remote event is detected, i.e. external event from external client or device is received, which changes the configuration mode, inactive, sleep or wake-up of the device**, [**0067, 0104-0105]: Ethernet message from endpoint in ethernet network includes header and at least one DU. Ethernet message further includes authentication information ASIL such as authentication key, a hash value, CRC, FRC and the like. The header includes destination address DA, source address SA, type filed, IP header, id, flag field, etc. [0129]: CAN message also includes control or management information. Based on these parameters from the endpoints, the encoding [such as convert or not] or transmitting [drop when authentication fails] can be performed by the gateway device**].

As per claim 31, Han discloses the method of claim 28, wherein encoding the message value in the second network data set comprises encapsulating the CAN message into the Ethernet message [**fig. 16, [0127-0131]].**

As per claim 32, Han discloses the method of claim 31, wherein encoding the message value further comprises processing at least one of a payload or a frame portion of the CAN message [f**ig. 16, [0127-0131]].**

As per claim 34, Han discloses the method of claim 28, wherein the at least one CAN message includes at least a first CAN message and a second CAN message [**fig. 16, fig. 13-fig. 15: multiple DU in each CAN message].**

As per claim 36, Han-Shearer discloses the system of claim 1, wherein the up-sampled data messages include the one or more messages and the one or more additional messages [**Shearer: claim 12, claim 22, [0058]], the one or more additional messages includes at least one of an interpolation or an extrapolation of original values of the one or more messages, and the up-sampled data messages both spanning data values of the one or more messages and non-spanning data values of the one or more messages [Shearer: claim 12, claim 22, claim 1 and claim 10], [0058], [0035], [0007]**]. Same rationale as in claim 1 applies.

As per claim 37, Han-Shearer discloses the system of claim 1, wherein the CND is structured to up-sample the one or more messages to match an up-sampling rate [**Shearer: [0007-0008], claim 12, 22 and claim 1]. Same rationale as in claim 1 applies.**

Claim(s) 27 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Han et al. (hereinafter Han, US 2018/0076970 A1**) in view of **Shearer et al. (US 2006/0018249 A1**) and further in view of **Brewton et al. (US 10,181,059 B1**).

As pe claim 27, Han-Shearer discloses the apparatus of claim 18 as set forth above.

However, Han-Shearer does not teach wherein the first network interface circuit, the translation circuit, and/or the second network interface circuit, are compliant with the Motor Industry Software Reliability Association standard and/or the AUTomotive Open System ARchitecture standard.

Brewton teaches wherein the first network interface circuit, the translation circuit, and/or the second network interface circuit, are compliant with the Motor Industry Software

Reliability Association standard and/or the AUTomotive Open System ARchitecture standard [col. 7 L55-65, col. 8 L22-28, fig. 2 – AUTOSAR compatible].

Therefore, it would have been obvious to a person of ordinary skilled in the art before the effective filing date of the claimed invention to modify Han-Shearer in view of Brewton in order to ensure the gateway device is compliant with Motor Industry Software Reliability Association standard or AUTomotive Open system Architecture standard.

One of ordinary skilled in the art would have been motivated in order ensure the gateway device is compatible based on the existing standards.

Claim(s) 35 is/are rejected under 35 U.S.C. 103 as being unpatentable over **Han et al. (hereinafter Han, US 2018/0076970 A1**) in view of **Shearer et al. (US 2006/0018249 A1**) and further in view of **O'Mahony et al. (US 2017/0134537 A1).**

As per claim 35, Han-Shearer discloses the system of claim 1 as set forth above.

However, Han-Shearer does not teach wherein the CND is further structured to down-sample one or more other messages between the first network of the first type and the second network of the second type, wherein the down-sample adjusts a number of the one or more other messages.

O'Mahony teaches CND is further structured to down-sample one or more other messages between the first network of the first type and the second network of the second type, wherein the down-sample adjusts a number of the one or more other messages [**[0079, 0109, 0003, 0099-0101]: decrease the data rate based on decrease in signal strength**].

Therefore, it would have been obvious to a person of ordinary skilled in the art before the effective filing date of the claimed invention to modify Han-Shearer in view of O'Mahony in order to down-sample one or more other messages between the first network of the first type and the second network of the second type, wherein the down-sample adjusts a number of the one or more other messages.

One of ordinary skilled in the art would have been motivated in order to keep data rate compatible with the destination network data rates, thus avoiding congestion or bottlenecks.

## *Conclusion*

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

- Rajan et al., US 2017/0072876 A1: Hardware-Accelerated protocol conversion in a automotive gateway controller.
- Chae et al., US 2019/0079842 A1 – Failure Diagnosis and Method for In-Vehicle Control Unit
- HEidari et al., US 6,512,739: Down Conversion within an X-DSL Receiver – Discloses Downstream packets may increase in size as a result of Interpolation.
- Tsuchinaga et al., US 2005/0238013 A1: Interpolation packet generated from preceding and subsequent packets by interpolation.

**THIS ACTION IS MADE FINAL.** Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to KAMAL B DIVECHA whose telephone number is **571-272-5863**. The examiner can normally be reached IFP Normal Hours M-F: 8am-4.30pm EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Colleen Fauz can be reached at 5712721667. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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